TECHNICAL TRAINING PROGRAM

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Training Days 2026 - A

CHAPTER 1 – WELCOME TO IT: SYSTEMS & INTERFACES

Let's Build the Foundations Together!

In this chapter, we will build the foundation needed to understand IT systems, computer hardware, software, operating systems, and the interfaces people use to interact with technology.

Before we begin, think about the following question:

Warm-Up Question

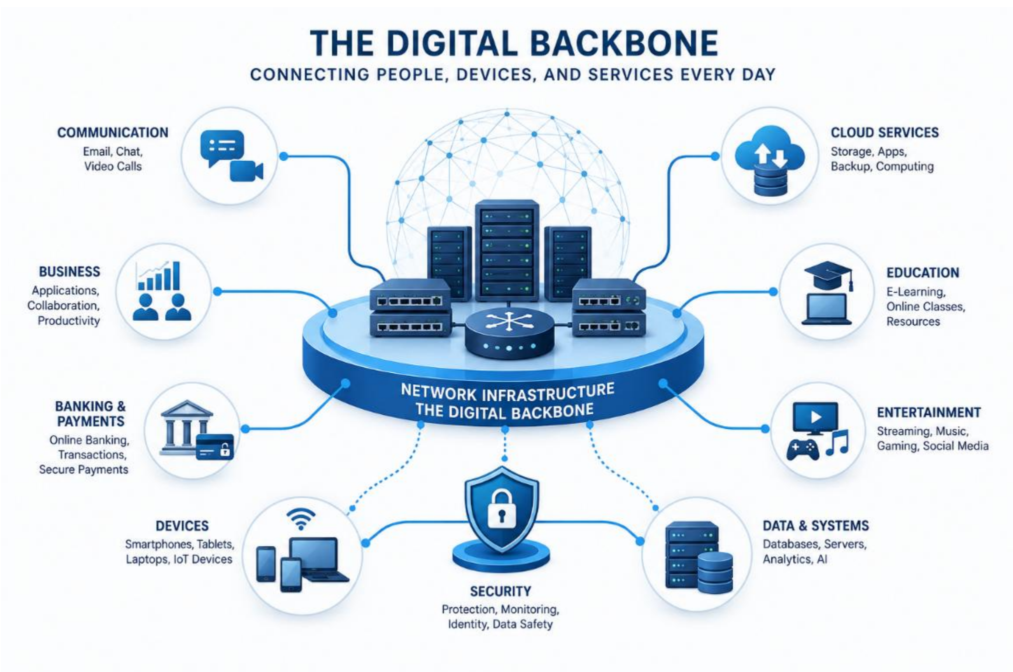
Imagine that your laptop, smartphone, and tablet suddenly stopped working.

Which part of your daily life would be affected first?

- Communication and email
- Online banking and payments
- Work or education
- Entertainment and streaming
- Gaming and social media

Most of the services we use every day depend on interconnected IT systems.

Information Technology is not simply a collection of computers and applications. It is the digital backbone that supports communication, business operations, education, healthcare, entertainment, and many other parts of modern life!



Our Digital Backbone

WHAT IS INFORMATION TECHNOLOGY?

Information Technology, commonly referred to as IT, is the use of computer systems, software, networks, and digital services to create, process, store, protect, and exchange information.

Information Technology

IT includes many different areas, such as:

- Computer hardware and operating systems
- Business applications and databases
- Network infrastructure and Internet services
- Cloud computing and virtualization
- Cybersecurity and data protection
- Technical support and system administration

IT enables organizations to automate processes, communicate efficiently, manage large amounts of data, and provide reliable services to employees and customers.

For example, IT allows a smartphone to synchronize email, a business application to access a database, and employees in different locations to collaborate using the same digital services.

Think About It

If the Internet is a highway, what would the vehicles represent?

The vehicles would represent data packets.

Just as vehicles carry people or goods from one place to another, packets carry data between devices, applications, and services across a network.

Throughout this training, we will examine the systems, devices, and technologies that make this communication possible.

A REAL-WORLD EXAMPLE: IT IN HEALTHCARE

To understand the importance of Information Technology, consider how a modern hospital operates:

Hospitals use electronic health record systems to store and manage patient information. Authorized doctors and nurses can quickly access medical histories, test results, allergies, and treatment plans. Having accurate information available at the correct time can improve both the quality and the safety of patient care.

IT systems are also used to:

- Schedule appointments and manage hospital resources
- Track medical supplies and equipment
- Send appointment reminders by email or SMS
- Connect laboratories, pharmacies, and medical departments
- Protect sensitive patient information
- Support remote consultations through telemedicine

Modern healthcare systems can also receive data from medical and wearable devices. This allows healthcare professionals to monitor important measurements and respond when unusual results are detected.

This example demonstrates an important principle:

IT is not only responsible for storing information. It connects people, devices, applications, and business processes so that an organization can operate effectively.

The same principle applies to banks, factories, schools, online shops, software companies, and almost every other modern organization.

IT in Medical Field

IT STRUCTURE

No two IT departments are exactly the same. Every organization has different business needs, technologies, risks, budgets, and numbers of users.

However, most IT departments follow the same basic principles: they support users, maintain systems, protect information, and ensure that technology remains available and reliable.

IT DEPARTMENT

Imagine entering the DTS offices at the beginning of a normal working day.

Employees are using their computers, accessing business applications, connecting to Wi-Fi, joining online meetings, and printing documents.

Who is responsible for keeping all these services working?

… WELCOME TO THE IT DEPARTMENT…

The Information Technology Department is responsible for managing and supporting an organization's technology environment.

Its responsibilities usually include:

- Computer systems and user devices
- Business applications and software
- Servers and storage systems
- Network and Internet connectivity
- User accounts and access permissions
- Data protection and information security
- Technical support and troubleshooting

The IT Department does not only react when something stops working. It also plans, monitors, maintains, and improves the systems that support the organization's daily operations.

What does the IT department actually do?

The main responsibilities of an IT Department can be grouped into several key areas

1. IT Management and Administration

- *Infrastructure Maintenance*: Ensuring that systems, devices, and networks remain operational and available.
- *User Support*: Assisting users with hardware, software, account, and access-related issues.
- *Employee Training*: Helping employees use IT systems safely and effectively.
- *System Update and Security*: Applying updates, reviewing security risks, and maintaining system protection.
- *Compliance and Licensing*: Managing software licenses and supporting compliance with organizational and legal requirements.

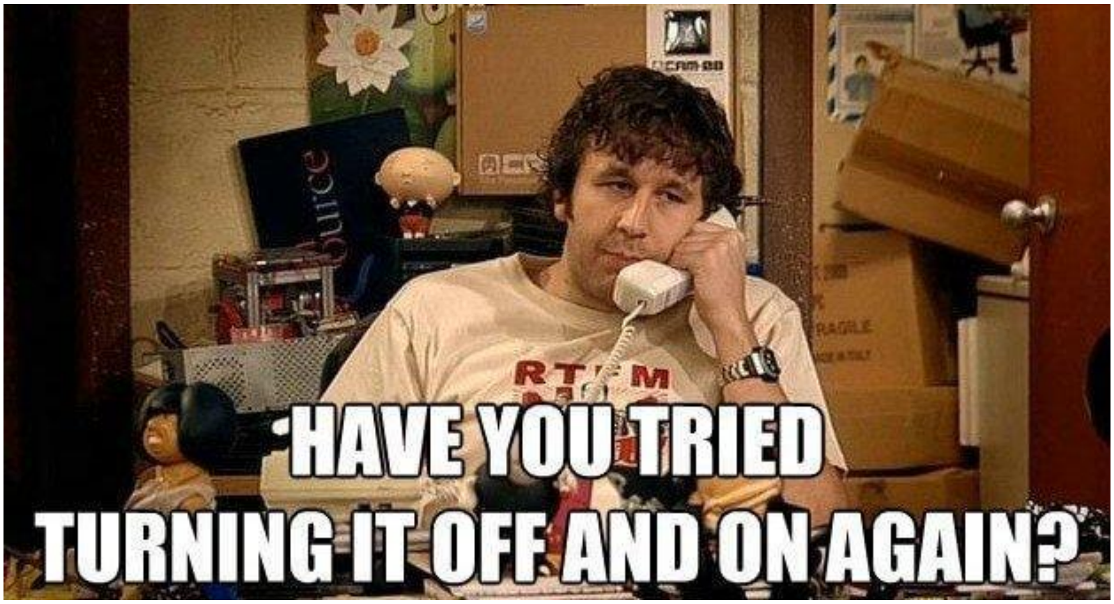
2. IT Procurement

    - *Requirements Analysis*: Identifying the organization’s hardware, software, licensing, and service requirements.
    - *Vendor & Solution Evaluation*: Comparing vendors and technical solutions based on cost, compatibility, security, support, and business needs.
    - *Acquisition & Deployment*: Purchasing approved solutions and coordinating their installation, configuration, and delivery.
    - *Asset Management*: Recording, tracking, and managing IT equipment and software throughout their lifecycle

3. IT Security
    - *Data Protection*: Protecting organizational information from unauthorized access, loss, alteration, or disclosure.
    - *Identity & Access Management*: Ensuring that users receive the correct level of access according to their role
    - *Threat Protection & Response*: Identifying suspicious activity and responding to security incidents.
    - *Security Awareness*: Helping employees recognize risks such as phishing, unsafe passwords, and social engineering.
    - *Security Updates & Controls*: Maintaining security tools, policies, patches, and protective configurations.

4. Technical Support
    - *Incident Resolution*: Investigating and resolving hardware, software, account, and connectivity issues.
    - *Service Requests*: Handling requests such as new accounts, software installations, equipment, or access permissions.
    - *User Assistance*: Guiding employees in the correct and safe use of systems and applications.
    - *Escalation*: Forwarding complex or high-impact issues to the appropriate specialist or support level.
    - *Documentation*: Recording symptoms, troubleshooting steps, solutions, and relevant technical information.

Do you know what the most popular catchphrase is in an IT department?



A Normal Day in an IT Dpt.

IT STRUCTURE

The IT Department is not simply a group of people waiting for something to break. It is a strategic business function with clearly defined responsibilities, specialized roles, and continuous communication between team members.

Think of it as a football team:

The IT Dpt. Football Team

- **Defense – Helpdesk**: Supports users and handles day-to-day technical issues.
- **Midfield - System Administrators**: Manage systems, servers, accounts, and services behind the scenes.
- **Forwards - Network Engineers**: Ensure that devices, networks, and services can communicate reliably.
- **Goalkeeper – Security Team**: Monitors risks, protects systems, and responds to security threats.

IT Department Main Roles	
Role	**Responsibilities**
Helpdesk Technician	First point of contact for user issues such as password resets, access problems, software issues, and device support
System Administrator	Manages servers, user accounts, system updates, backups, and core IT services
Network Administrator	Manages switches, routers, LAN/WAN connectivity, wireless networks, and network monitoring
Security Analyst	Monitors security tools, investigates threats, supports incident response, and helps protect systems and data
Cloud/Infrastructure Engineer	Builds and maintains virtual machines, cloud platforms, infrastructure services, and backup solutions
IT Manager	Oversees the IT team, prioritizes projects, manages resources, and reports to senior management

Information: In smaller organizations, one person may perform several of these roles, while larger organizations usually divide them among specialized teams!

IT PROCESSES

Like every other department in an organization, the IT Department does not operate without structure or planning!

Imagine working in IT without clear processes.

One user reports, “I cannot print.” Another says, “The server is unavailable.” Someone else requests a new user account.

Without a structured way to record, prioritize, assign, and resolve these requests, the IT Department would quickly become disorganized.

Without IT Processes

IT processes help teams organize their work and provide consistent, traceable, and reliable services. These activities can be grouped into three key areas:

1. **Incident and Service Request Management**

An **incident** is an unexpected interruption or reduction in the quality of an IT service, such as a printer failure, a network outage, or an application that stops working.

A **service request** is a standard user request, such as creating a new account, installing approved software, or granting access to a shared resource.

Incidents and service requests are recorded in a ticketing system so that they can be categorized, prioritized, assigned, tracked, and documented until completion. Many organizations use ticketing platforms, such as ***Jira Service Management*** or similar service desk systems, to manage this work.

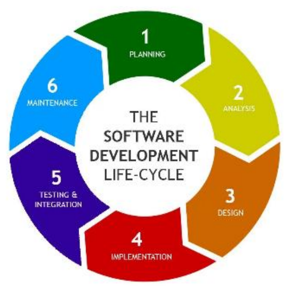
2. **IT Operations**
    - Applying updates and security patches
    - Checking backups and recovery results
    - Monitoring systems, services, and network performance
    - Managing configurations and scheduled maintenance
    - Reviewing alerts, logs, and system health

Each and every day we have to follow these routines to keep the IT infrastructure alive.

3. **Software Development**

When an organization develops or customizes software, the work usually follows a structured process known as the **Software Development Life Cycle**, or **SDLC**.

The SDLC helps teams plan, build, test, deploy, and maintain software in a controlled and organized way.



Software Development Life Cycle

a. **Planning**: Defining the purpose, scope, resources, and expected outcome.
b. **Requirements**: Identifying what users and the organization need from the software.
c. **Design**: Planning the structure, interface, data, and technical solution.
d. **Development**: Writing and integrating the software code.
e. **Testing**: Checking that the software works correctly and meets the requirements.
f. **Deployment**: Releasing the software to its intended users or environment.
g. **Maintenance**: Correcting issues, applying updates, and improving the software over time.

Question:

Why should IT teams document even a small action, such as restarting a user's computer?

Answer:

Documentation creates a record of what happened, what actions were taken, and whether the issue was resolved. It also helps other team members identify repeated problems and continue the investigation if the issue returns.

IT INFRASTRUCTURE

Nowadays, almost every organization relies on its IT infrastructure to operate effectively. IT infrastructure is the combination of hardware, software, networks, facilities, and services required to support users, business processes, communication, data, and digital operations.

For example, when an employee signs into a workstation, accesses a shared file, joins an online meeting, or uses a business application, several infrastructure components work together to provide the service.

IT INFRASTRUCTURE TYPES

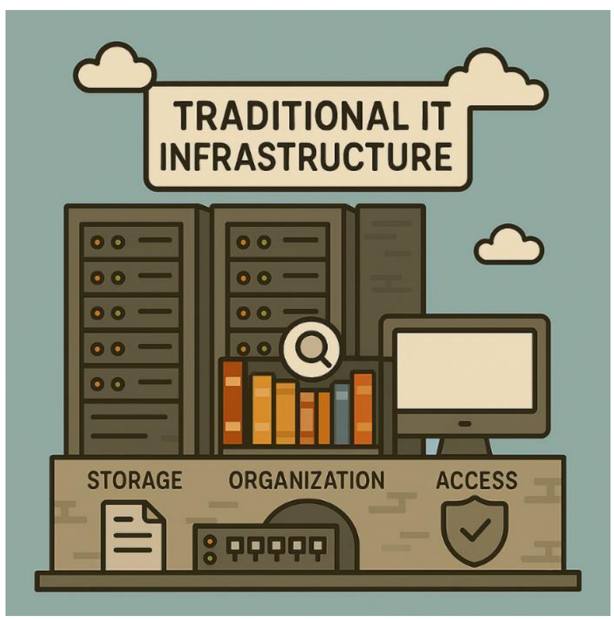
IT infrastructure can be implemented in different ways. The most common approaches are **traditional on-premises** infrastructure, **cloud-based** infrastructure, and a combination of both, known as **hybrid** infrastructure.

Each approach has different advantages, limitations, costs, management requirements, and security responsibilities. We will begin with traditional on-premises infrastructure.

A. TRADITIONAL ON-PREMISES IT INFRASTRUCTURE

Traditional on-premises IT infrastructure consists of hardware, software, network devices, storage systems, and supporting facilities that are installed and managed at the organization's own premises.

The organization is responsible for purchasing, configuring, maintaining, securing, and replacing the infrastructure throughout its lifecycle.



Traditional IT Infrastructure

IT INFRASTRUCTURE ELEMENTS

Now, let’s examine the main elements that make up an IT infrastructure.

Hardware

Hardware includes the physical devices and equipment used to support users, applications, data, communication, and business operations.

- Desktop computers and laptops
- Tablets and smartphones
- Servers and storage systems
- Routers, switches, and wireless access points
- Printers and other peripheral devices
- Racks, power systems, cooling, and supporting facilities
- Network and power cabling

Peripherals: Peripherals are devices that extend the functionality of a computer system, such as monitors, keyboards, mice, printers, scanners, webcams, and external storage devices.

Software

Software includes the operating systems, applications, platforms, and management tools used to operate the infrastructure and provide services to employees, customers, and business partners.

We usually encounter the following:

- Operating systems
- Business applications
- Web and application servers
- Database management systems
- Ticketing and service-management platforms, such as Jira Service Management
- Customer Relationship Management systems
- Enterprise Resource Planning systems
- Building Management Systems
- Security, monitoring, backup, and management tools

Network:

Network infrastructure connects users, devices, applications, and services, allowing them to communicate and exchange data securely and reliably:

- Routers and switches
- Wireless access points
- Firewalls and security appliances
- Network interfaces and adapters
- Copper and fiber-optic cabling
- Internet and WAN connections
- Network services, such as DNS and DHCP

<table>
<tr><th colspan="2">Traditional IT Infrastructure</th></tr>
<tr><th>Pros</th><th>Cons</th></tr>
<tr><td><u>Greater Control</u>: The organization has direct control over hardware, software, configurations, and data location</td><td><u>High Initial Cost</u>: The organization must purchase hardware, software licenses, facilities, and supporting equipment</td></tr>
<tr><td><u>Customization</u>: Systems can be designed and configured according to specific technical and business requirements</td><td><u>Maintenance Responsibility</u>: Internal staff or contracted specialists must manage updates, repairs, backups, monitoring, and security</td></tr>
<tr><td><u>Local Access</u>: Internal services may continue to operate even if the Internet connection is unavailable</td><td><u>Limited Scalability</u>: Expanding capacity may require purchasing and installing additional equipment</td></tr>
<tr><td rowspan="2"><u>Direct Management</u>: The organization can choose its own hardware, software, security tools, and maintenance procedures</td><td><u>Facility Requirements</u>: Servers and network equipment may require suitable power, cooling, space, and physical security</td></tr>
<tr><td><u>Hardware Lifecycle</u>: Equipment must eventually be upgraded, replaced, or safely disposed of</td></tr>
</table>

B. CLOUD-BASED IT INFRASTRUCTURE

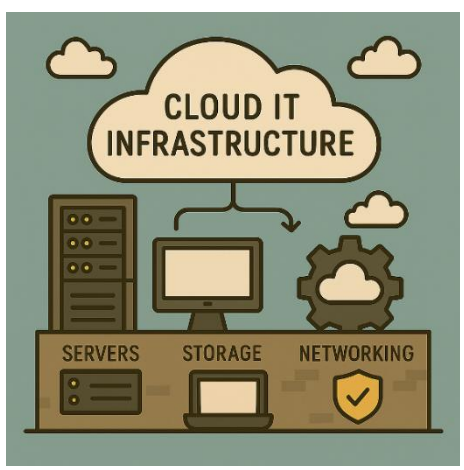
Cloud-based infrastructure provides computing resources, applications, storage, and services through a cloud provider's data centers. These resources are accessed through a network connection and can usually be provisioned or adjusted according to the organization's requirements.

The cloud provider manages parts of the underlying infrastructure, but the customer remains responsible for areas such as user accounts, access permissions, data protection, configurations, and the secure use of cloud services.

Cloud-Based IT Infrastructure	
Pros	**Cons**
Scalability: Resources can often be increased or reduced according to demand	**Dependence on Connectivity**: Access to cloud services usually depends on reliable network and Internet connections
Flexibility: Services can be accessed from different locations and devices when properly authorized	**Ongoing Costs**: Subscription and usage-based charges must be monitored and controlled
Reduced Initial Hardware Investment: Organizations may avoid purchasing and maintaining large amounts of physical infrastructure	**Shared Responsibility**: The customer remains responsible for identities, permissions, data, configurations, and secure usage
Rapid Deployment: New services and resources can often be provisioned quickly	**Reduced Physical Control**: The customer does not directly manage the provider's physical infrastructure
Provider-Managed Infrastructure: The provider maintains the underlying data-center facilities and physical hardware	**Provider Dependency**: Service availability, supported features, locations, and contractual conditions depend partly on the provider

C. Hybrid Infrastructure

Many organizations use a hybrid approach, combining on-premises systems with cloud services. For example, an organization may keep certain servers locally while using cloud-based email, backups, collaboration tools, or business applications.



Cloud-Based Infrastructure

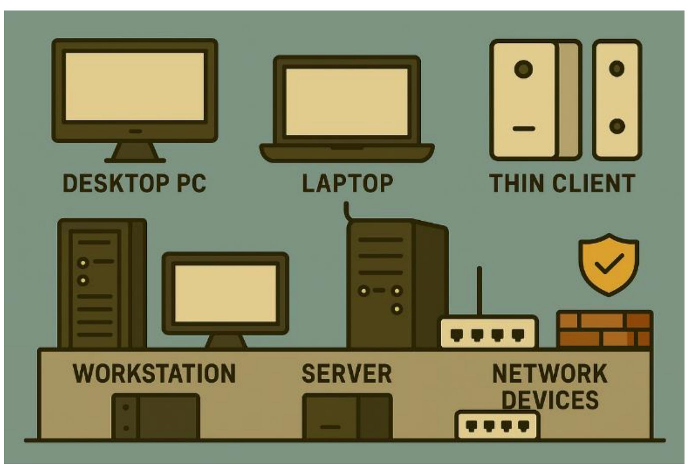
IT INFRASTRUCTURE IN DETAILS

TRADITIONAL ON-PREMISES HARDWARE INFRASTRUCTURE

Hardware provides the physical foundation on which IT systems and services operate. In an on-premises environment, the organization owns or directly manages the devices and equipment used by employees, applications, networks, and business services.

Common hardware devices include:

- Desktop Computer: A personal computer designed to remain at a fixed location, commonly used in offices, schools, and homes.
- Laptops: A portable computer that combines a display, keyboard, battery, storage, and processing components in a single device.
- Thin Client: A lightweight endpoint designed to access applications, desktops, or computing resources hosted on a central server or cloud platform. Thin clients usually have fewer local resources and are commonly used with Virtual Desktop Infrastructure or Remote Desktop Services.
- Workstation: A high-performance computer designed for demanding professional workloads, such as engineering, software development, scientific analysis, video production, or 3D design.
- Network Devices: Devices that connect systems and control the data traffic across a network. Common examples include switches, routers, wireless access points, and firewalls. Hubs may still be encountered in older environments but are rarely used in modern networks.
- Server: A physical or virtual system that provides resources, applications, data, or network services to other devices known as clients. Servers are commonly identified by the services they provide, such as file servers, web servers, DNS servers, or database servers.
- Mobile Devices: Smartphones and tablets that allow users to access communication tools, business applications, cloud services, and organizational data while working from different locations.

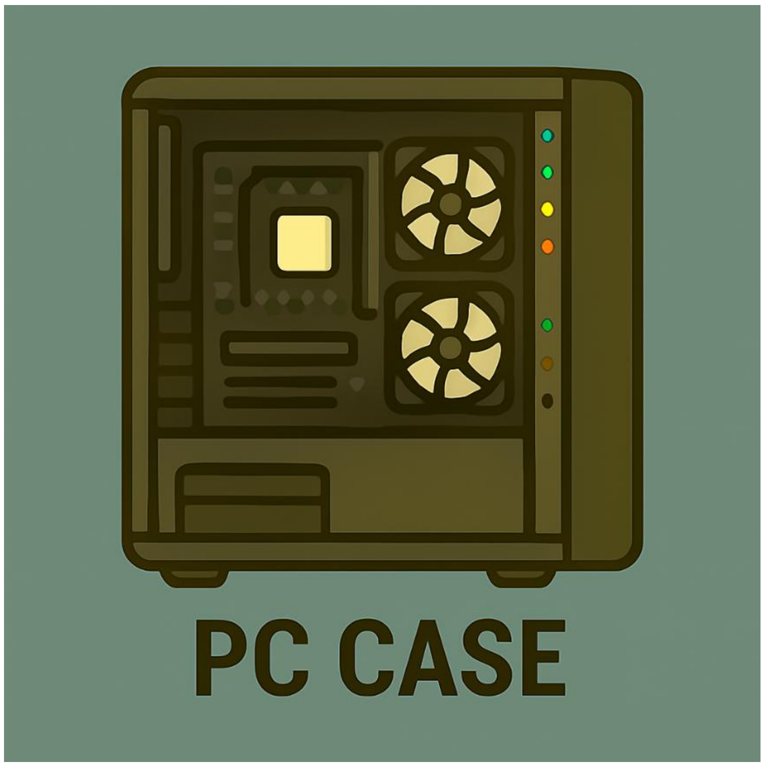


Traditional Hardware IT Infrastructure

THE MAIN COMPONENTS OF A COMPUTER

PC Case: The enclosure that contains and protects the internal components of a computer. It provides physical protection, supports airflow and cooling, and allows the components to be securely installed and connected.

Information: *PC cases are available in different sizes and form factors. The selected case must be compatible with the motherboard, power supply, cooling system, storage devices, and expansion cards*!



<u>Motherboard</u>: The main circuit board of a computer. It connects and allows communication between the processor, memory, storage devices, expansion cards, power supply, and external peripherals. A motherboard is designed according to specific standards and must be compatible with the selected processor, memory, case, power supply, storage interfaces, and expansion devices.

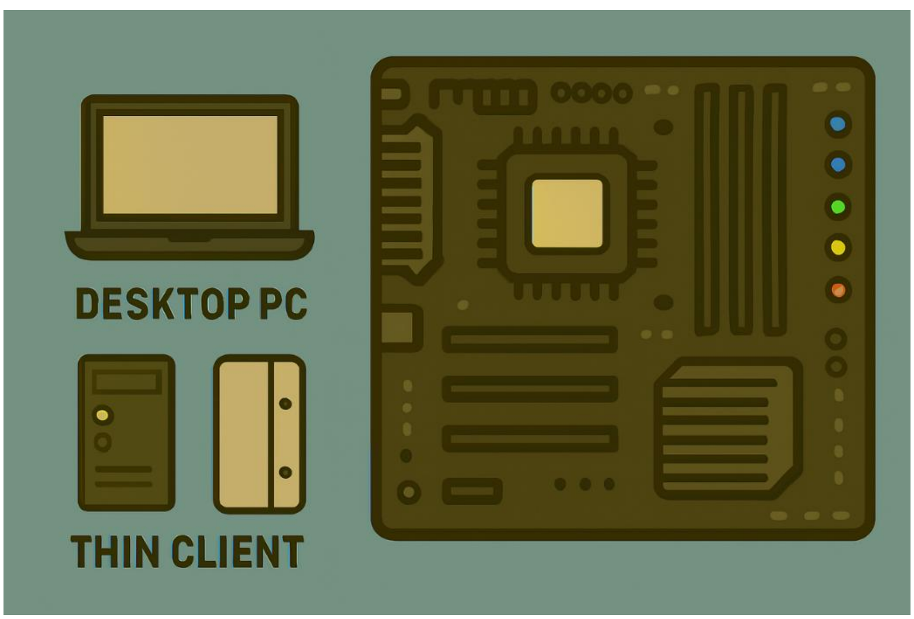
Important motherboard components and connections include:

- CPU socket
- Memory slots
- Storage connectors
- PCI Express expansion slots
- Power connectors
- Firmware chip containing UEFI
- Integrated network, audio, and input/output controllers
- External ports, such as USB, Ethernet, audio, and display connections

In modern systems, many functions that were previously handled by separate motherboard chips are now integrated into the processor or managed by a single chipset.

Historical Note*: Older motherboard designs commonly used separate Northbridge and Southbridge chips. Modern systems integrate many of these functions into the processor and a unified chipset.*

A processor must match the motherboard's CPU socket, chipset support, firmware compatibility, and manufacturer platform. An AMD processor cannot be installed in a motherboard designed for an Intel processor, and vice versa.



CPU (*Central Processing Unit*): The primary processor of a computer. It executes instructions, performs calculations, processes data, and coordinates many of the operations required by the operating system and applications.

The CPU receives instructions from memory, interprets them, performs the required operation, and stores or forwards the result. Its performance depends on several factors, including its architecture, number of cores, clock speed, cache memory, power limits, and the workload being processed.

The CPU is often described as the brain of the computer because it processes instructions and coordinates operations.

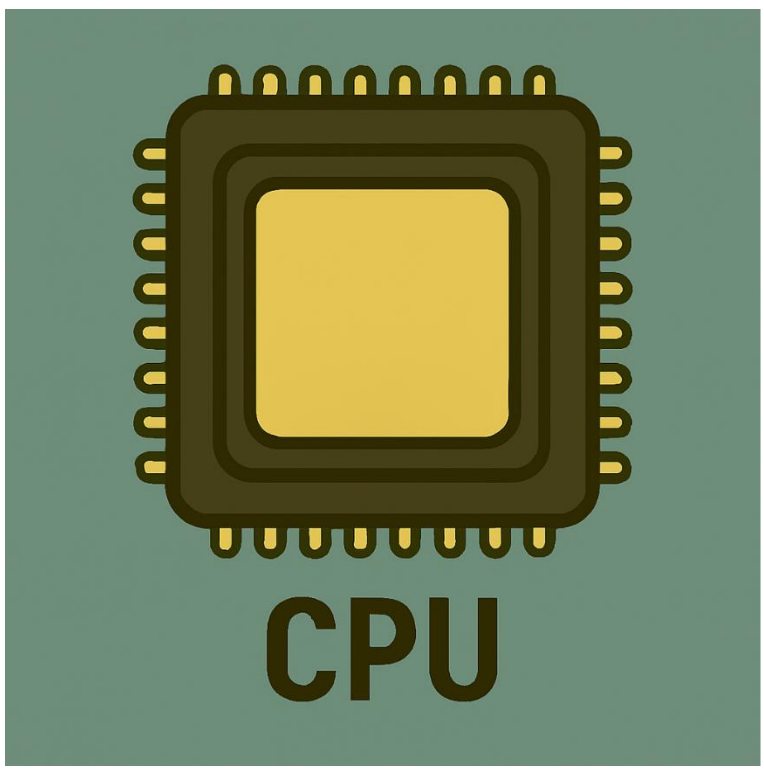
Important CPU characteristics include:

Cores: Independent processing units within the CPU that allow multiple tasks to be handled more efficiently.

Threads: Logical execution paths that allow a core to manage more than one sequence of instructions.

Clock Speed: The rate at which the CPU performs processing cycles, usually measured in gigahertz.

Cache Memory: Small, high-speed memory located close to or inside the CPU, used to store frequently required data and instructions.



The CPU contains several important functional components that work together to process instructions and data.

Arithmetic Logic Unit (ALU): Performs arithmetic calculations and logical operations, such as comparisons.

Floating-Point Unit (FPU): Performs calculations involving decimal and floating-point numbers.

Control Unit: Coordinates the execution of instructions and controls the flow of data between the CPU, memory, and other components.

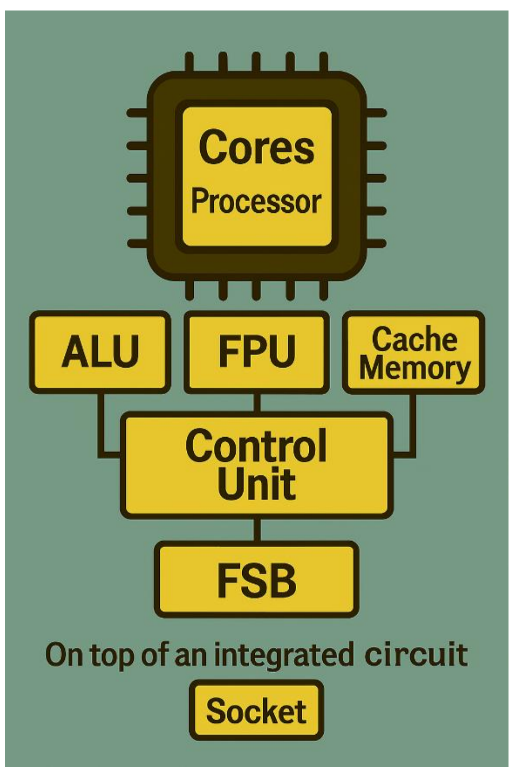
Registers: Very small, high-speed storage locations inside the CPU that temporarily hold instructions, addresses, and data currently being processed.

Cache Memory: High-speed memory located inside or very close to the CPU, used to reduce the time required to access frequently used data and instructions.

Modern processors usually contain **multiple cores**. Each core can execute instructions independently, allowing the computer to process several tasks more efficiently.

The CPU communicates with memory, storage controllers, expansion devices, and the motherboard chipset through high-speed internal connections and interfaces.

In older computer architectures, the processor communicated with the motherboard through a connection known as the Front Side Bus (FSB). In modern systems, many memory and communication functions are integrated directly into the processor, and newer high-speed interconnects are used instead.



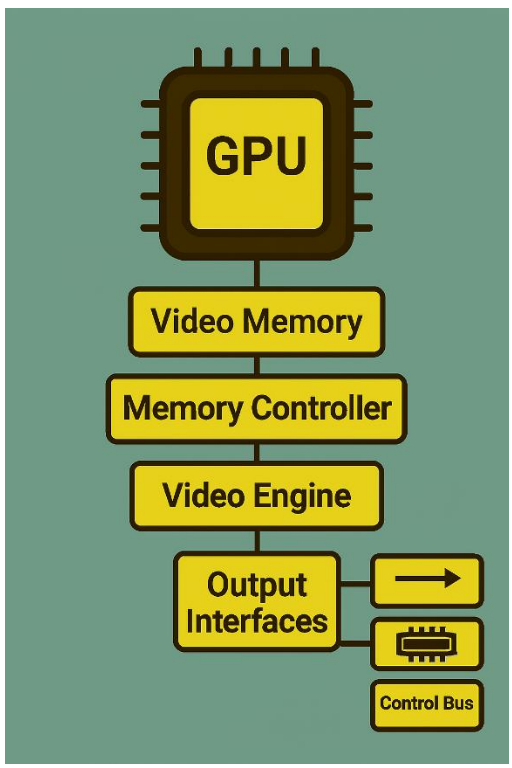
GPU (Graphics Processing Unit): A processor designed to handle graphics, image processing, video, and other highly parallel workloads.

The GPU is responsible for generating the visual output displayed on a monitor. It can also accelerate workloads such as video editing, three-dimensional rendering, scientific calculations, machine learning, and certain software-development tasks.

A GPU may be implemented in two main ways.

- Integrated GPU: Built into the processor or system architecture and usually shares the computer's main memory.
- Dedicated GPU: A separate graphics processor, commonly installed on an expansion card, with its own video memory and cooling system.

Dedicated GPUs can consume significant power and generate considerable heat. The computer must therefore have suitable power delivery, airflow, and cooling.

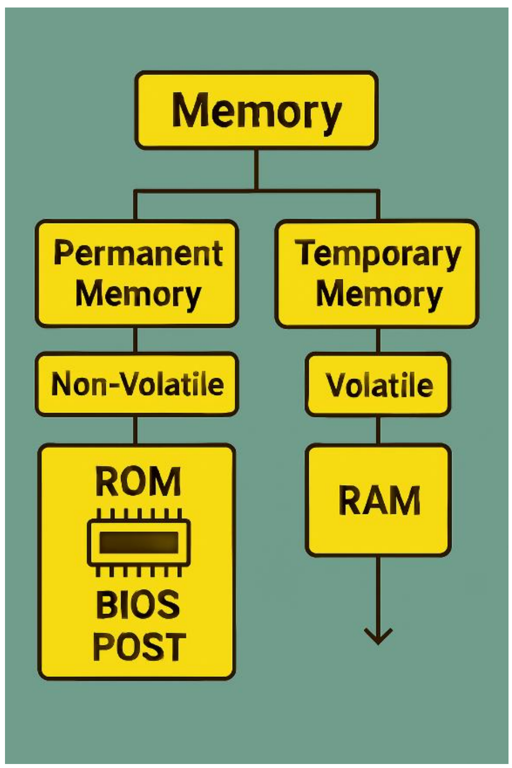


Memory: Computer memory can be divided into two main categories: volatile memory and non-volatile memory.

- Volatile Memory: Requires electrical power to retain its contents. The main example is RAM (Random Access Memory), which temporarily stores the data and instructions currently required by the operating system and applications. When the computer is powered off, the contents of RAM are lost.
- Non-Volatile Memory: Retains its contents even when electrical power is removed. Examples include flash memory, solid-state storage, and the memory chip used to store the motherboard's UEFI firmware.

The amount and speed of RAM can affect system performance, especially when several applications or demanding workloads are running at the same time.

Note: *Storage devices, such as HDDs and SSDs, are also non-volatile, but they are used for long-term data storage rather than as the computer's working memory*.



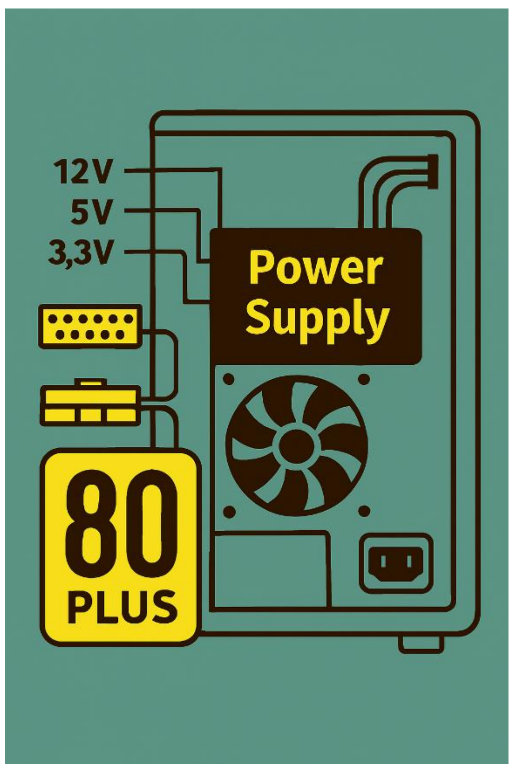
PSU - Power Supply Unit: The component that converts alternating current from the electrical outlet into the regulated direct-current voltages required by the computer's internal components.

The power supply must provide sufficient capacity for the processor, graphics card, storage devices, cooling system, and other connected components. It must also be compatible with the computer case, motherboard, and required power connectors.

When selecting a PSU, important factors include:

- Required power capacity
- Electrical efficiency
- Component quality and safety protections
- Correct connectors and form factor
- Reliability and manufacturer support

80 PLUS is an efficiency certification that indicates how efficiently a power supply converts electrical power under specific test conditions. It is useful when comparing power supplies, but it does not by itself guarantee overall quality, reliability, or suitability for a particular computer.



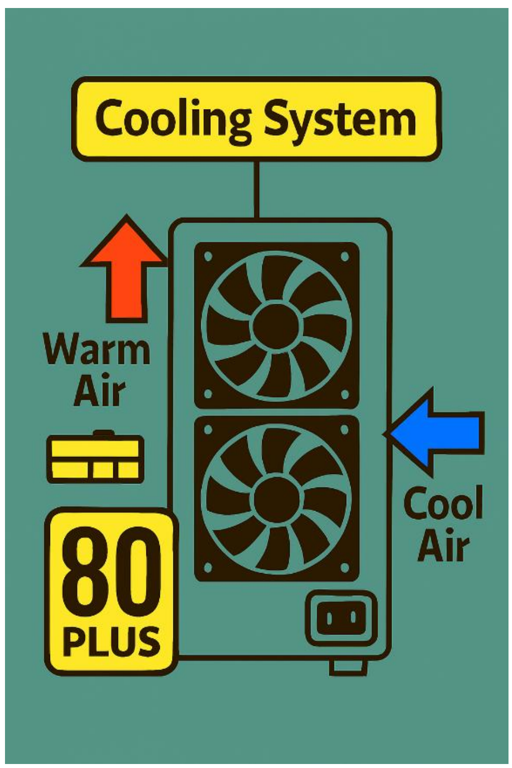
Cooling Systems: Computer components consume electrical power and generate heat while operating. A suitable cooling system is required to keep temperatures within safe operating limits and maintain system stability and performance.

Case fans create airflow by drawing cooler air into the computer and exhausting warmer air from the case. The processor and graphics card normally use dedicated heatsinks and fans, while some systems may use liquid cooling.

Effective cooling depends on:

- Correct airflow direction
- Clean air vents and filters
- Properly installed heatsinks and fans
- Suitable thermal interface material
- Adequate space around the computer
- Regular removal of dust

If temperatures become too high, the system may reduce performance through thermal throttling, become unstable, shut down unexpectedly, or experience a shorter component lifespan.



Network Interface Card (NIC): The hardware component that allows a computer, server, or other device to connect to a network and exchange data with other systems.

A NIC may provide:

- Wired Ethernet connectivity
- Wireless network connectivity
- One or more physical network ports
- A unique hardware address known as a MAC address
- Support for specific network speeds and standards

Network interfaces may be integrated into the motherboard or installed separately as expansion cards or external adapters.

The supported speed depends on the device, network standard, cabling, switch, and overall network design. Common environments may use connections ranging from standard desktop Ethernet to high-speed server and data-center networking.

A network connection can only operate at the highest speed supported by all relevant components, including the NIC, cable, switch port, and network configuration.

Note: *A device may have more than one network interface. Servers often use multiple interfaces for redundancy, performance, management, or separation of network traffic*.